

# Лабораторная работа №07

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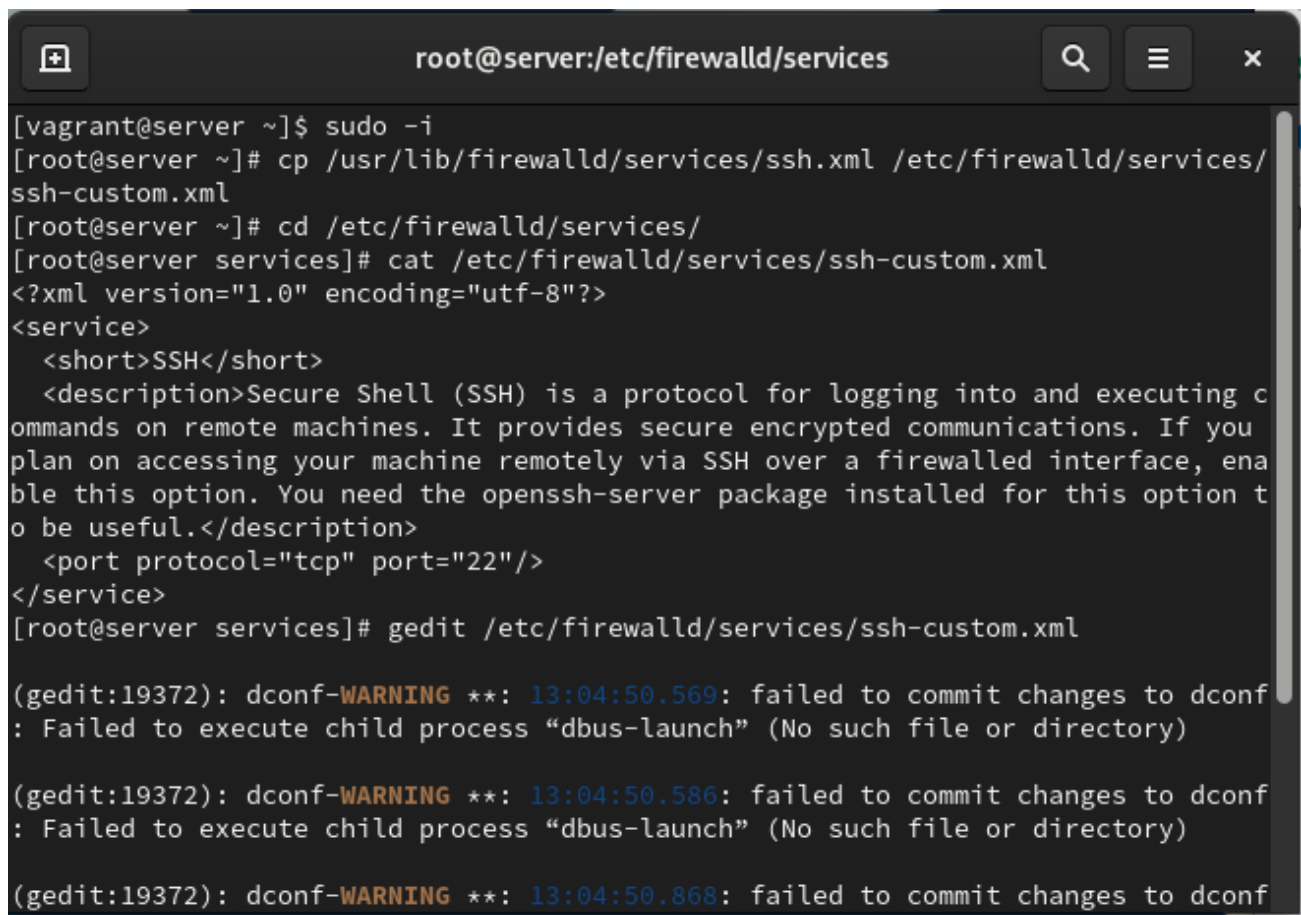
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### Цель работы

Приобретение практических навыков администрирования сетевых подсистем

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**На основе существующего файла описания службы ssh создадим файл с собственным описанием. Перейдём в каталог служб и посмотрим содержимое созданного файла (рис. @fig-1):**



```
root@server:/etc/firewalld/services

[vagrant@server ~]$ sudo -i
[root@server ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server ~]# cd /etc/firewalld/services/
[root@server services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing c
ommands on remote machines. It provides secure encrypted communications. If you
plan on accessing your machine remotely via SSH over a firewalled interface, ena
ble this option. You need the openssh-server package installed for this option t
o be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
[root@server services]# gedit /etc/firewalld/services/ssh-custom.xml

(gedit:19372): dconf-WARNING **: 13:04:50.569: failed to commit changes to dconf
: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:19372): dconf-WARNING **: 13:04:50.586: failed to commit changes to dconf
: Failed to execute child process "dbus-launch" (No such file or directory)

(gedit:19372): dconf-WARNING **: 13:04:50.868: failed to commit changes to dconf
```

Рис. 1: Этап 1

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Откроем файл описания службы на редактирование и заменим порт 22 на новый порт (2022). В этом же файле скорректируем описание службы для демонстрации, что это модифицированный файл службы (рис. @fig-2):



Рис. 2: Этап 2

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Просмотрим список доступных Firewalld служб, чтобы убедиться, что наша новая служба ещё не отображается (рис. @fig-3):

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Перезагрузим правила межсетевого экрана с сохранением информации о состоянии и вновь выведем на экран список служб, а также список активных служб. Убедимся, что созданная нами служба отображается в списке доступных для Firewalld служб, но не активирована (рис. @fig-4):

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Добавим новую службу в Firewalld и выведем на экран список активных служб, чтобы убедиться в её активации (рис. @fig-5):

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Организуем на сервере переадресацию с порта 2022 на порт 22 (рис. @fig-6):

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На клиенте попробуем получить доступ по SSH к серверу через порт 2022 (рис. @fig-7):

---

На сервере посмотрим, активирована ли в ядре системы возможность перенаправления IPv4-пакетов (рис. @fig-8):

```
root@server:/etc/firewalld/services

t-encoding not supported

** (gedit:19372): WARNING **: 13:08:59.634: Set document metadata failed: Setting attribute metadata::gedi
t-position not supported

(gedit:19372): dconf-WARNING **: 13:08:59.642: failed to commit changes to dconf: Failed to execute child
process "dbus-launch" (No such file or directory)
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisa
pp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitco
in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
app-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
ips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
ssdp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls teln
et tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server warpinator wbem-htt
p wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdm
cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]#
```

Рис. 3: Этап 3

```
root@server:/etc/firewalld/services

cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --reload
bash: firewall-cmd: command not found...
[root@server services]# firewall-cmd --reload
success
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisa
pp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitco
in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
app-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
ips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog sysl
og-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server warpinat
or wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman
wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]#
```

Рис. 4: Этап 4

```
root@server:/etc/firewalld/services

cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --reload
bash: firewall-cmd: command not found...
[root@server services]# firewall-cmd --reload
success
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisa
op2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitco
in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
lps slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog sysl
og-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsd vnc-server warpinat
or wbm-http wbm-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman
wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]#
```

Рис. 5: Этап 5

```
wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]# firewall-cmd --add-service=ssh-custom
success
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server services]# firewall-cmd --add-service=ssh-custom --permanent
success
[root@server services]# firewall-cmd --reload
success
[root@server services]#
```

Рис. 6: Этап 6

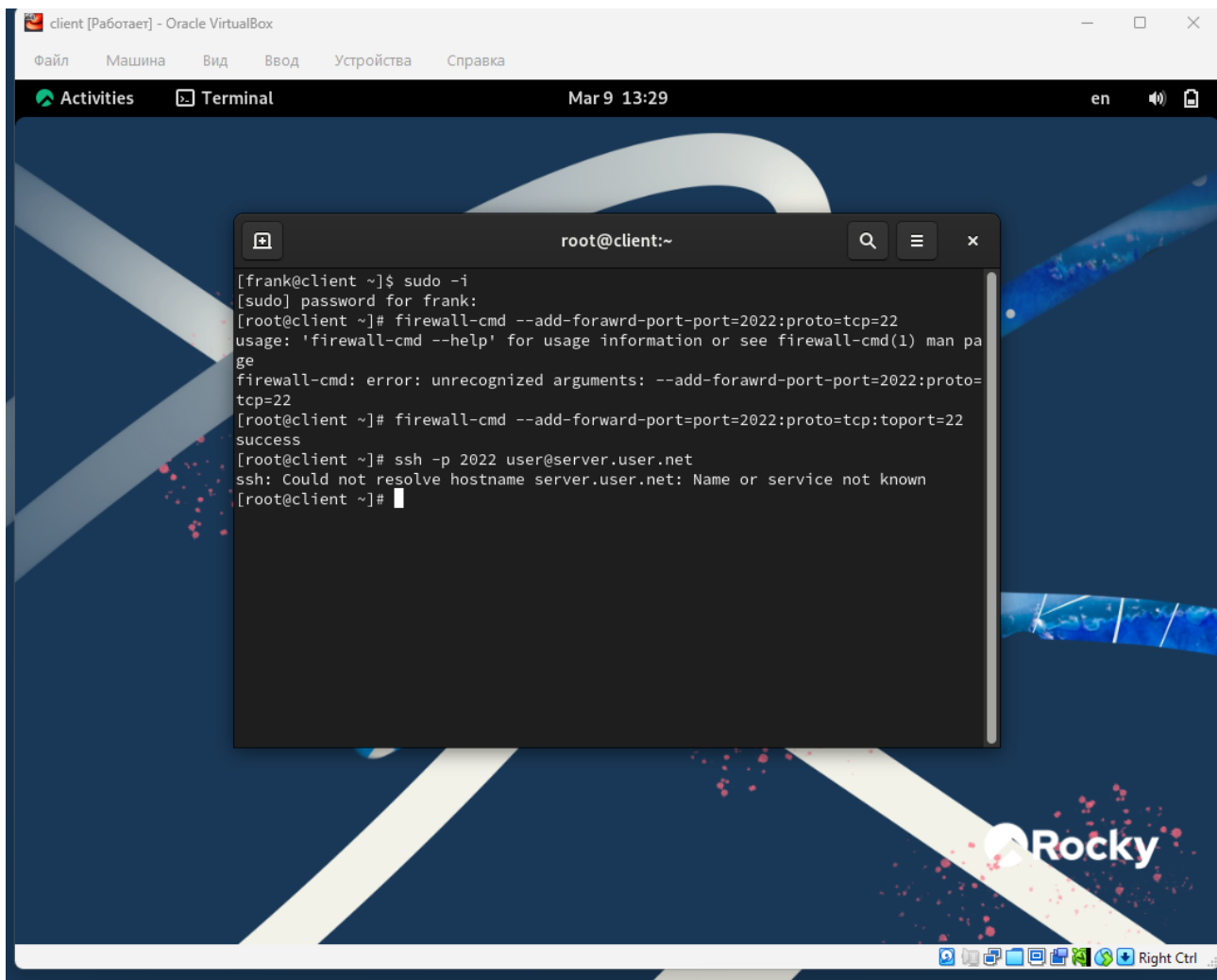


Рис. 7: Этап 7

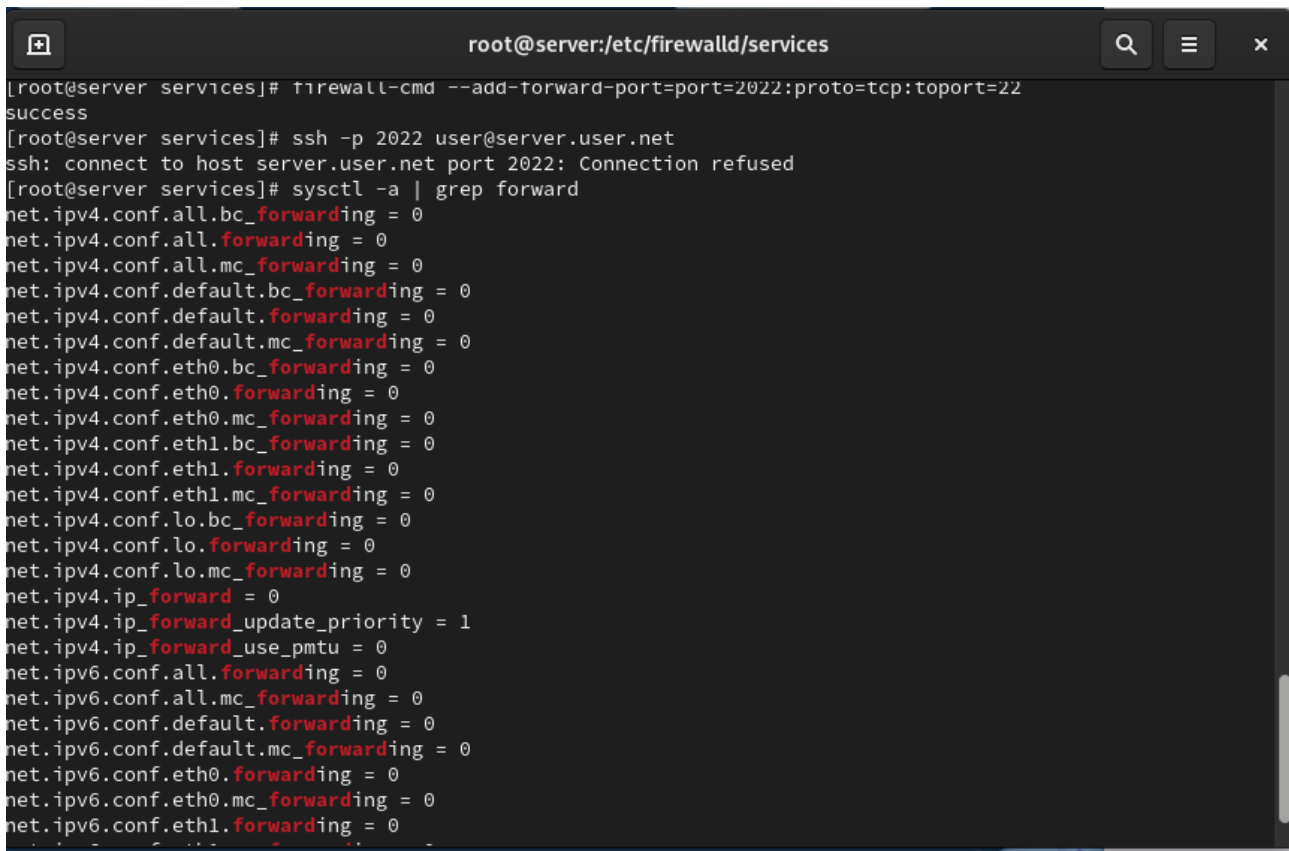


Рис. 8: Этап 8

**Включим перенаправление IPv4-пакетов на сервере и затем включим маскардинг (рис. @fig-9):**

```
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server services]# firewall-cmd --zone=public --add-masquerade --permanent
success
[root@server services]# firewall-cmd --reload
success
[root@server services]#
```

Рис. 9: Этап 9

---

**На клиенте проверим доступность выхода в Интернет после настройки маскардинга (рис. @fig-10):**

---

**На виртуальной машине server перейдём в каталог для внесения изменений в настройки внутреннего окружения, создадим в нём каталог firewall, в который поместим в соответствующие подкаталоги конфигурационные файлы FirewallD. В каталоге создадим файл firewall.sh (рис. @fig-11):**

---

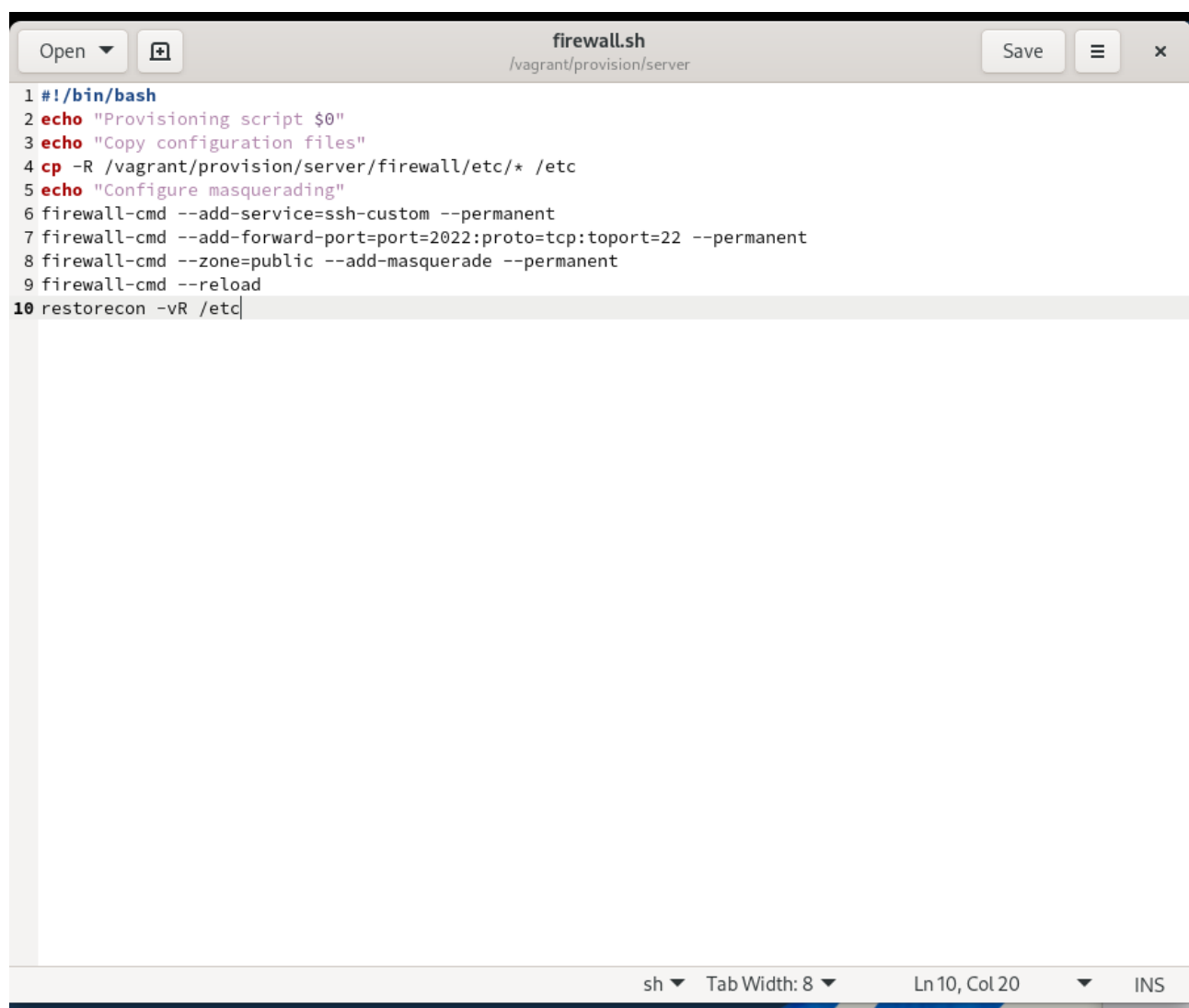
## Выводы

В ходе выполнения лабораторной работы были приобретены практические навыки

---

**Спасибо за внимание!**

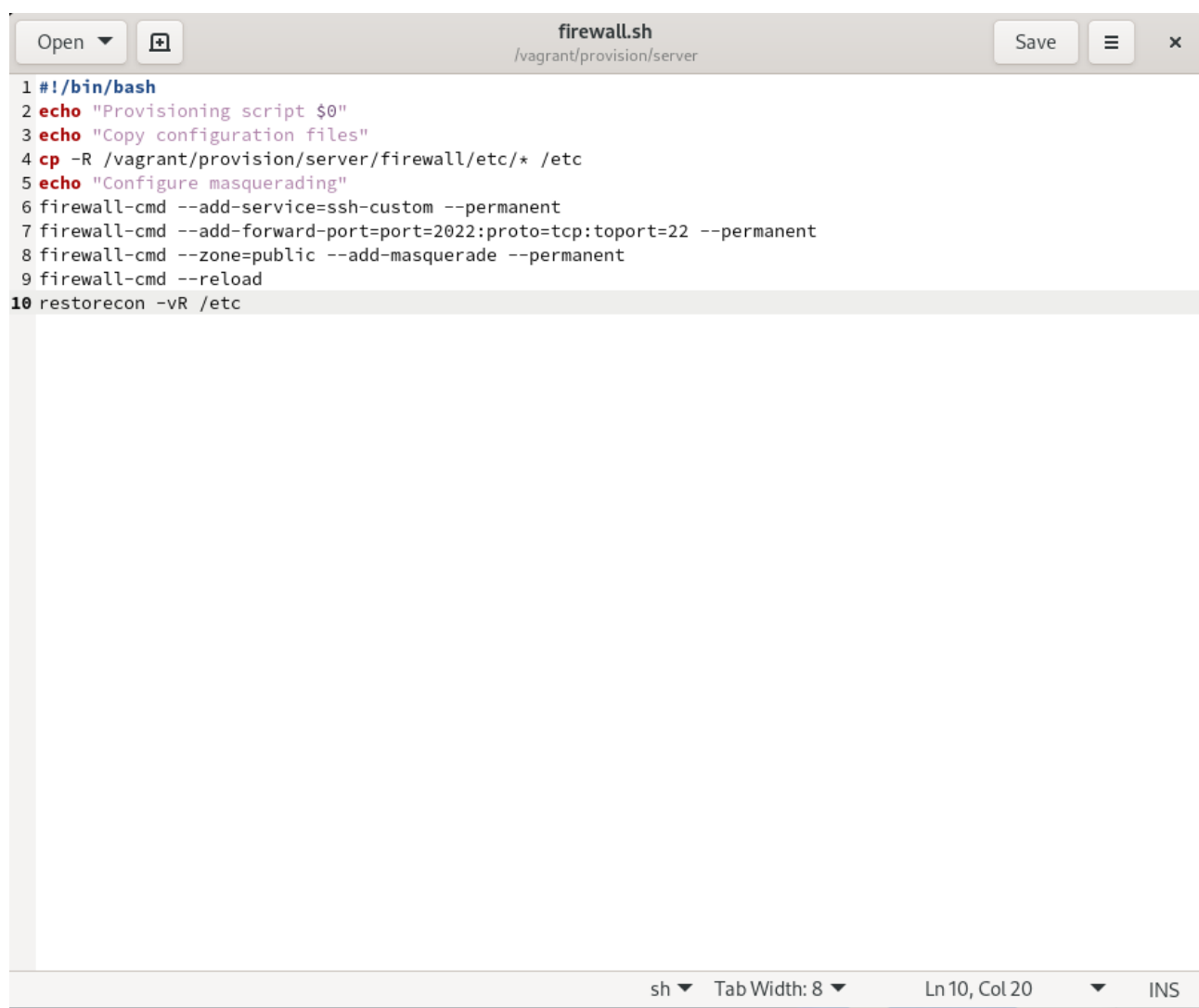
**Вопросы?**



The image shows a code editor window with the title 'firewall.sh' and the file path '/vagrant/provision/server'. The editor contains a shell script with 10 lines. Line 10 is highlighted, showing the command 'restorecon -vR /etc|'. The status bar at the bottom indicates 'sh', 'Tab Width: 8', 'Ln 10, Col 20', and 'INS'.

```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Copy configuration files"
4 cp -R /vagrant/provision/server/firewall/etc/* /etc
5 echo "Configure masquerading"
6 firewall-cmd --add-service=ssh-custom --permanent
7 firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
8 firewall-cmd --zone=public --add-masquerade --permanent
9 firewall-cmd --reload
10 restorecon -vR /etc|
```

Рис. 10: Этап 10



```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Copy configuration files"
4 cp -R /vagrant/provision/server/firewall/etc/* /etc
5 echo "Configure masquerading"
6 firewall-cmd --add-service=ssh-custom --permanent
7 firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
8 firewall-cmd --zone=public --add-masquerade --permanent
9 firewall-cmd --reload
10 restorecon -vR /etc
```

Рис. 11: Этап 11